

Rational expressions



1 Fully simplify the following rational expressions:

a $\frac{49v}{35}$

b $\frac{15k}{20k}$

c $\frac{70v}{710}$

d $\frac{42xy}{18y}$

e $\frac{18p^7}{9p^3}$

f $\frac{30mn^2}{9m^2n}$

g $\frac{8k^2}{-12k}$

h $\frac{-12xy}{48x^2}$

2 For each of the following rational expressions:

i Factor the numerator and denominator.

ii Identify the greatest common factor between the numerator and denominator.

iii Fully simplify the rational expression.

a $\frac{4a}{4ab - 20ac}$

b $\frac{2fg - 6fh}{2f}$

c $\frac{6r + 3rst}{3rs - 12rt}$

d $\frac{4xy + 8xy^2}{12x^3y^3}$

3 Fully simplify the following rational expressions:

a $\frac{6x - 16}{12}$

b $\frac{x - 4}{4 - x}$

c $\frac{3y - 45}{15 - y}$

d $\frac{3x - 12}{9x + 15}$

e $\frac{3uv - 12uw}{3u}$

f $\frac{4u}{4uv - 20uw}$

g $\frac{2x + 10y}{x + 5y}$

h $\frac{3x - 15y}{x - 5y}$

4 Fully simplify the following rational expressions:

a $\frac{a^2 - 81}{9 - a}$

b $\frac{x - 5}{x^2 - 25}$

c $\frac{x - 6}{x^2 - 36}$

d $\frac{m^2 - 1}{1 + m}$

e $\frac{m^2 - 100}{m - 10}$

f $\frac{50m^2 + 70mn}{80m^2}$

g $\frac{x - 1}{x^2 - 2x + 1}$

h $\frac{x - 2}{x^2 + 4x - 12}$

i $\frac{6(x + 12)}{5x^2 + 120x + 720}$

j $\frac{x^2 + 7x + 10}{x + 2}$

k $\frac{x^2 + 4x + 4}{10(x + 2)}$

l $\frac{2x^2 + 4x - 198}{5(x - 9)}$

5 Fully simplify the following rational expressions:

a $\frac{2r - 8}{r^2 - 16}$

b $\frac{2x + 8}{x^2 + 4x}$

c $\frac{20 - 5x}{2x^2 - 8x}$

d $\frac{15x - 10}{3x^3 - 2x^2}$

e $\frac{12r + 60}{3r^2 - 75}$

f $\frac{6(p - q)^2}{6p^2 - 6q^2}$

g $\frac{15 - 3xy}{x^2y - 5x}$

h $\frac{12m^2 + 6mn}{18mn + 36m^2}$

6 Fully simplify the following rational expressions



a $\frac{x^2 + 8x + 16}{(x - 5)(x + 4)}$

b $\frac{(x - 7)(x - 1)}{x^2 - 9x + 8}$

c $\frac{8x^2 + 2x - 15}{(4x - 5)^2}$

d $\frac{(3x + 2)^2}{18x^2 - 3x - 10}$

e $\frac{x^2 - 14x + 49}{x^2 - 49}$

f $\frac{x^2 + x - 12}{x^2 - 3x}$

g $\frac{a^2 + 10a + 24}{a^2 - 16}$

h $\frac{p^2 + 31p + 220}{p^2 - 121}$

i $\frac{4x^2 + 12x + 9}{9 - 4x^2}$

j $\frac{4x^2 - 3x - 10}{4 - x^2}$

k $\frac{x^2 + 2x - 8}{8 - 2x^2}$

l $\frac{x^2 - 2x - 8}{x^3 - 16x}$

7 Fully simplify the following rational expressions:

a $\frac{2x^2 + 10x - 100}{2x + 20}$

b $\frac{9x^2 + 9x}{10x^2 + 20x + 10}$

c $\frac{x^2 + 14x + 49}{x^2 + 4x - 21}$

d $\frac{x^2 - 10x + 25}{x^2 + 5x - 50}$

e $\frac{4x^2 + 8x - 32}{x^2 + 3x - 10}$

f $\frac{x^3 - x^2 - 20x}{x^2 + 7x + 12}$

g $\frac{25 - 16x^2}{4x^2 + 7x - 15}$

h $\frac{x^2 + x - 12}{5x^2 + 18x - 8}$

i $\frac{8x^2 - 18x + 10}{4x^2 + x - 5}$

j $\frac{2x^2 - 7x + 3}{4x^2 - 17x + 15}$

k $\frac{2x^2 - 13x + 15}{2x^2 - x - 3}$

l $\frac{4x^2 + 7x - 2}{6 + x - x^2}$

8 Luciana has attempted to fully simplify a rational expression and showed her work:

1 $\frac{2x + 7}{2x^2 + 6x} = \frac{2x + 7}{2x(x + 3)}$ Factor the denominator

2 $= \frac{7}{x + 3}$ Divide out $2x$ from the numerator and denominator

Identify where Luciana has made an error and explain what it is.

9 Qin has attempted to fully simplify a rational expression and showed his work:

1 $\frac{8x^2 - 24x}{x^2 + 5x - 24} = \frac{8x(x - 3)}{(x + 8)(x - 3)}$ Factor the numerator and denominator

2 $= \frac{8x}{x + 8}$ Divide out $(x + 3)$ from the numerator and denominator

3 $= \frac{8x}{x} + \frac{8x}{8}$ Additive property of fractions

4 $= 8 + x$ Simplify the fractions

Identify where Qin has made an error and explain what it is.

- 10 For each of the following, find the value(s) of  that would allow the rational expression to be simplified into a linear expression:

a $\frac{kx^2 + 12x}{4x}$

b $\frac{x^2 + kx + 42}{x + 6}$

c $\frac{9x^2 + 6x + k}{3x}$

d $\frac{x^2 - 7x - 18}{x + k}$